https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	R is the independent variable and *t* is the dependent variable **or** vary *R* and measure *t*	**1**
	keep the number of turns on the coil/*N* constant	**1**
	Methods of data collection	
	labelled diagram or correct symbols including: • labelled (d.c.) power supply • switch in series with power supply, resistor and coil • complete workable circuit	**1**
	circuit diagram to measure *R*, e.g. ammeter and voltmeter correctly positioned **or** *R* connected to ohmmeter with no other connections (**not** ohmmeter in main circuit)	**1**
	method to determine *t* (of a few milliseconds) e.g. use (storage) oscilloscope **or** current/voltage sensor connected to datalogger/computer	**1**
	method to determine *A*, e.g. micrometer/calipers to determine diameter of coil **and** $A = \pi d^2 / 4$	**1**
	Method of analysis	
	plot a graph of *t* against 1 / *R* (allow log *t* against log *R*)	**1**
	relationship valid if a straight line passing through the origin is produced (allow gradient = –1 for graph of log *t* against log *R*)	**1**
	$K = \dfrac{\text{gradient} \times L}{AN^2}$.	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 open switch/switch off (high voltage) circuit before changing the resistor/touching components **or** ensure no bare wires/use shrouded connectors	
	D2 wear (insulating) gloves to prevent electric shock/electrocution	
	D3 keep *A* and *L* constant	
	D4 use ruler/calipers to measure *L*	
	D5 repeat measurements of diameter in different directions/at points along the coil and average	
	D6 method to determine *R* e.g. $R = V / I$ linked to correct circuit diagram for ammeter/voltmeter method **or** measure resistance using ohmmeter	
	D7 repeat experiment for each value of *R* and average *t*	
	D8 method to determine *t*: use of time-base from oscilloscope explained **or** use of time axis of output from data logger/computer explained	
	D9 use smaller values of *R* to increase *I*	
	D10 reduce *L* or increase *N* or increase *A* to increase *t*	

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